# LUKE KIMBALL

**Applied AI Engineer — OpenRouter-Native Multi-Model Routing · Agent Harnesses · Build-Over-Buy Internal Tools**

559-789-7214 · kimballluke@gmail.com · Provo, UT — Remote (US) · linkedin.com/in/lukekimball2789 · github.com/duketopceo · luke-the-duke.com

## PROFILE

OpenRouter-native operator-builder who ships internal agentic tooling instead of buying software built for the masses. Multi-model daily driver: routes frontier models to map a repo, smaller models to decode, large models for implementation, and small models to organize — each task to the cheapest capable model. Forked Agent Zero into a personal, OpenRouter-native harness, built the eval and spend tooling around it, and designed a tenant-aware agent platform with security and evals built in. Support/GTM fluency from sales, ticketing, CRM, and ops.

## CORE CAPABILITIES

- **Multi-Model Routing & Cost** — OpenRouter as primary router, with routing, fallbacks, budgets, and spend tracking treated as first-class. Hands-on with local open-weight inference via Ollama and vLLM.
- **Evals & Model Quality** — Built *orchestral*, an eval harness testing whether a cheap orchestrator plus cheap workers matches frontier output, and *Argus*, a vision-model E2E harness with hard budget caps. Turns routing and cost tradeoffs into a measurable number.
- **Agent Harnesses** — Claude Code, Cursor, Agent Zero, Hermes, Gemini; hands-on with Factory AI, OpenCode, Grok, Perplexity Computer. Forked Agent Zero into "Khan"; designed Pace's agentrun pipeline (TierGate, tool registry, approval gates).
- **Agentic Tooling & LLM Infra** — Rust/Axum gateway + Go/Chi agent backend + Python/LangGraph reasoning; Supabase Postgres 17 with RLS tenant isolation; Redis Streams event bus; Stripe metering; Stytch B2B SSO.
- **Build-Over-Buy Internal Tools** — Wraps open-source tools (Snipe-IT, MeshCentral, Vaultwarden) into single internal products; built a helpdesk dashboard and web-app pitch decks instead of buying SaaS or making PDFs.
- **Security, Evals & Reliability** — Outbound allowlist (SSRF prevention), approval gates, RLS isolation, auth-gated internal APIs, secrets in Vaultwarden. Owns the unglamorous last 20%: CI gates before merge, incident runbooks, cross-tenant isolation enforcement, Sentry + structlog observability; Stratum ships a 334+ test suite.

## SELECTED AI & INFRASTRUCTURE PROJECTS

**orchestral** github.com/duketopceo/orchestral

Python · OpenRouter · evals · E2B · sandboxing · CI

- OpenRouter eval harness: does a cheap orchestrator plus cheap workers produce frontier-quality output? Grew an E2B-compatible isolated code runtime so evals have somewhere safe to execute.
- Provider-agnostic — runs BYOK against any OpenRouter model; 417 commits, CI-gated.

**Argus** github.com/duketopceo/Argus

TypeScript · OpenRouter · vision models · E2E testing · GitHub Actions

- Open-source vision-model E2E testing harness — record, replay, heal, assert. BYOK through OpenRouter with hard budget caps, self-hosted runner, and GitHub PR reporting. The open TestDriver alternative.
- Budget caps, failure recovery, and instrumentation treated as part of the interface, not an afterthought.

**Pace Server (Pace HQ)** pacehq.io

Rust · Axum · Go · Chi · Python · LangGraph · React · Supabase · Postgres 17 · RLS · Redis · Stripe · Stytch

- Private-cloud OS for SMBs — wraps open-source tools (chat, vault, e-sign, files) into one product with one login and one bill.
- Designed the tenant-aware agent system: 8-step agentrun pipeline, TierGate autonomy tiers (1–5), tool registry, audit trail, approval gates, outbound allowlist (SSRF prevention); model routing via OpenRouter. Platform live (gateway, auth, Stripe billing); real tool execution and Python reasoning service still in progress.

**Khan — Agent Zero fork**

Agent Zero · OpenRouter · plugins · multi-agent · Mac app

- Private product fork of Agent Zero — a personal Linux-backed agent harness with model/harness freedom. OpenRouter-native, plugin system, multi-agent delegation, secrets management, browser DOM annotation, Mac app distribution.
- Strategy, design, and governance live in-repo; one-command install ships a PATH CLI plus a Mac app wrapper.

**wisp** github.com/duketopceo/wisp

Python · whisper.cpp · Hyprland · local inference · computer use

- Voice computer-use agent for the Omarchy desktop: push-to-talk → whisper.cpp → local decision model → guarded hyprctl launch, with a breathing dim overlay while it listens.
- Local-first — the decision model runs on the machine, no round trip for the hot path; every action the agent may take is gated.

**Kurultai** github.com/duketopceo/kurultai

Rust · SQLite · FTS5 · sqlite-vec · MCP · axum

- Local knowledge brain for agents — indexes notes, chats, repos, and docs into one SQLite store with FTS5 full-text + sqlite-vec vector search, exposed through an MCP server.
- One store for retrieval instead of whole-vault dumps — search/ask/MCP return excerpts with citations.

**Stratum Engine** stratumhq.app

FastAPI · Supabase · Fly.io · Stripe · Retell · voice AI

- Secondary production voice-agent project — 24/7 Retell agent for contractors (missed-call recovery, follow-ups); 334+ tests.

## EXPERIENCE

**IT Support & Data Specialist — Bartlett Roofing** 2025 – Present · UT / ID

- Promoted within 4 months; scope grew to IT support and data work across UT + ID for a ~200-person roofing org.
- Containerized services (Snipe-IT, MeshCentral, Vaultwarden) on bare-metal Linux; built a helpdesk dashboard and web-app pitch decks instead of buying SaaS or making PDFs.

- Built read/write internal APIs for the IT team, gated behind authentication; Python automations and a Next.js dashboard cut manual follow-up and reporting.
- Asked to train the IT team on AI tooling in month one — runs Claude Code, Cursor, and OpenRouter routing daily across the org.

**Data Intern / Manufacturing IT Liaison — LiveView Technologies** 2024 – 2025 · UT

- Cross-team liaison coordinating between engineering and production teams; automated data-pipeline fixes in Python.
- Built Tableau dashboards for production analytics — defect rates, throughput, and cycle times.

**Inside Sales & Trainer — Vivint Smart Home** 2020 – 2024 · UT

- Power-ranked top 6 in department for 4 consecutive months on commission-heavy, no-base comp; trained 30+ new hires.

## EDUCATION

**Certificate, Data Analytics — MTECH Vocational** 2024 – 2025 · **Computer Engineering & Economics — BYU-Idaho** 3 semesters